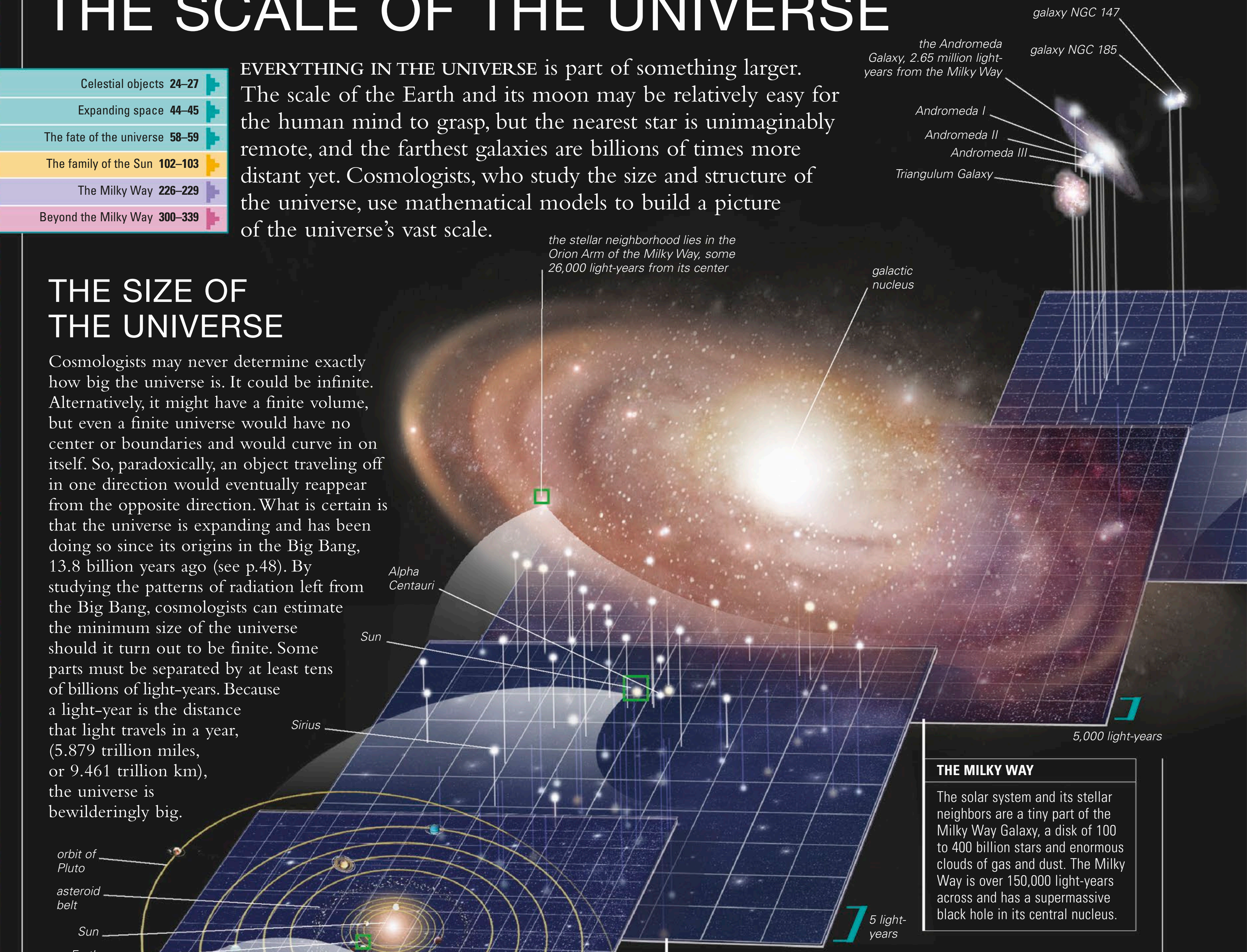
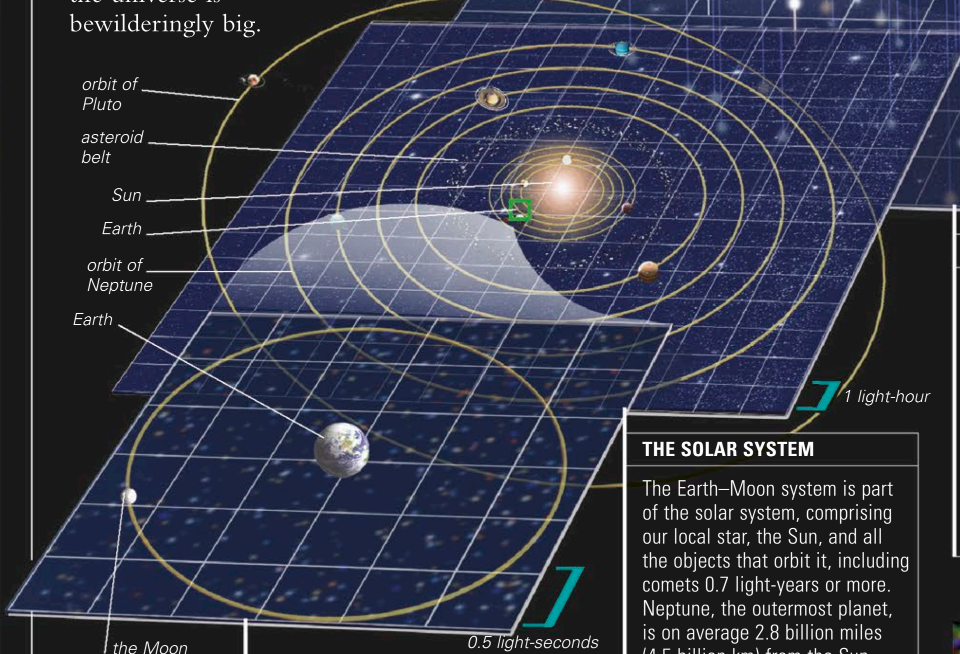
THE SCALE OF THE UNIVERSE

EVERYTHING IN THE UNIVERSE is part of something larger. The scale of the Earth and its moon may be relatively easy for the human mind to grasp, but the nearest star is unimaginably remote, and the farthest galaxies are billions of times more distant yet. Cosmologists, who study the size and structure of the universe, use mathematical models to build a picture of the universe's vast scale.

THE SIZE OF THE UNIVERSE

Cosmologists may never determine exactly how big the universe is. It could be infinite. Alternatively, it might have a finite volume, but even a finite universe would have no center or boundaries and would curve in on itself. So, paradoxically, an object traveling off in one direction would eventually reappear from the opposite direction. What is certain is that the universe is expanding and has been doing so since its origins in the Big Bang, 13.8 billion years ago (see p.48). By studying the patterns of radiation left from the Big Bang, cosmologists can estimate the minimum size of the universe should it turn out to be finite. Some parts must be separated by at least tens of billions of light-years. Because a light-year is the distance that light travels in a year, (5.879 trillion miles, or 9.461 trillion km), the universe is bewilderingly big.

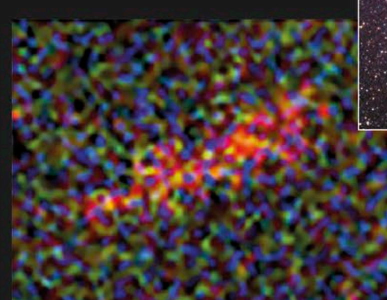



THE MILKY WAY
The solar system and its stellar neighbors are a tiny part of the Milky Way Galaxy, a disk of 100 to 400 billion stars and enormous clouds of gas and dust. The Milky Way is over 150,000 light-years across and has a supermassive black hole in its central nucleus.

THE STELLAR NEIGHBORHOOD
The closest star system to the Sun, Alpha Centauri, lies 4.37 light-years, or 26 trillion miles (41 trillion km) away. Within 16 light-years of the Sun are 59 stars. These include 13 star systems, each containing two or three stars. An example is Sirius, the brightest star in the night sky, which is actually a system of two stars of markedly different size.

THE SOLAR SYSTEM
The Earth–Moon system is part of the solar system, comprising our local star, the Sun, and all the objects that orbit it, including comets 0.7 light-years or more. Neptune, the outermost planet, is on average 2.8 billion miles (4.5 billion km) from the Sun.

THE EARTH AND MOON
Earth has an average diameter of 7,920 miles (12,740 km), while the diameter of the Moon's orbit around Earth is about 480,000 miles (770,000 km). A space probe sent to the Moon typically takes just a few days to get there.



REMOTE OBJECT
This embryonic galaxy, called SPT0615-JD (right), is one of the most distant known. The light from it, detected by the Hubble and Spitzer Space Telescopes, began its journey toward us 13.3 billion years ago.

VIEW FROM EARTH
The Milky Way Galaxy is a complex 3-D structure, but from our position within it, it appears as a 2-D band across the sky (above).